

Homework 1

$$1. a \begin{pmatrix} & 0 & 1 & 2 \\ 0.85 & 0.11 & 0.04 \\ 0.10 & 0.84 & 0.06 \\ 0 & 0 & 1 \end{pmatrix}$$

want: ${}_3P_X^{12}$

states	t=0	t=1	t=2	t=3	probabilities	joint probability
	1	→ 1	→ 1	→ 2	$(0.84)^2(0.06)$	0.042336
	1	→ 1	→ 2	→ 2	$(0.84)(0.06)$	0.050400
	1	→ 2	→ 2	→ 2	0.06	0.060000
	1	→ 1	→ 0	→ 2	$(0.84)(0.10)(0.04)$	0.003360
	1	→ 0	→ 2	→ 2	$(0.10)(0.04)$	0.004000
	1	→ 0	→ 1	→ 2	$(0.10)(0.11)(0.06)$	0.000760
	1	→ 0	→ 0	→ 2	$(0.10)(0.85)(0.04)$	0.003400

$${}_3P_X^{12} = 0.164156$$

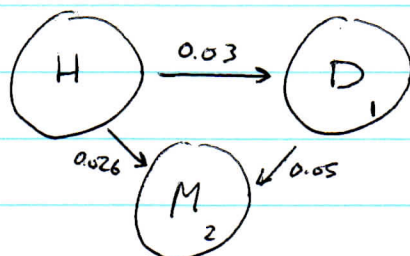
b.

states	t=0	t=1	t=2	joint probability
	0	→ 0	→ 1	$0.85 \times 0.11 = 0.0935$
	0	→ 1	→ 1	$0.11 \times 0.84 = 0.0924$
				0.1859

want: stocks from 0, exactly 3 of 6 in 2 years

$$\Rightarrow \binom{6}{3} (0.1859)^3 (1 - 0.1859)^3 = 0.06933$$

2.



$$\begin{aligned}
 {}_6P_x^{01} &= \int_0^6 e^{-(0.03+0.026)t} (0.03) e^{-0.05(6-t)} dt \\
 &= 0.03 \int_0^6 e^{-0.3-(0.056-0.05)t} dt = 0.03 e^{-0.3} \int_0^6 e^{-0.006t} dt \\
 &= \frac{0.03}{0.006} e^{-0.3} (1 - e^{-0.006 \times 6}) dt = 0.1310
 \end{aligned}$$

$${}_6P_x^{\infty} = e^{-\int_0^6 (0.026+0.03) dt} = e^{-0.056 \times 6} = 0.7146$$

$${}_6P_x^{11} = e^{-\int_0^6 0.05 dt} = e^{-0.05 \times 6} = 0.7408$$

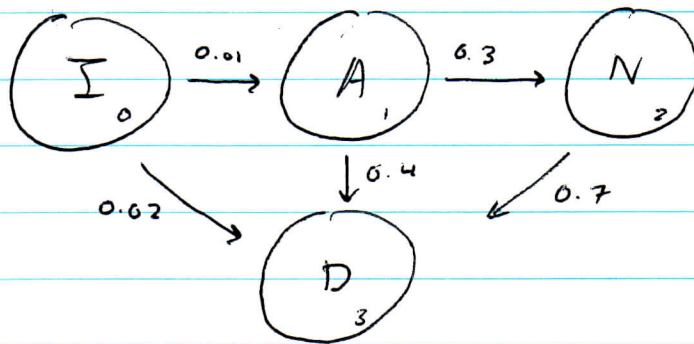
$${}_6P_x^{12} = \int_0^6 e^{-0.05t} (0.05)(1) dt = \left(\frac{0.05}{0.05} \right) (1 - e^{-0.05 \times 6}) = 0.2592$$

$$\begin{aligned}
 {}_6P_x^{02} &= \int_0^6 e^{-(0.026+0.03)t} (0.026)(1) dt + \int_0^6 e^{-(0.026+0.03)t} (0.03)(1 - e^{-0.05(6-t)}) dt \\
 &= \frac{0.026}{0.056} (1 - e^{-0.056 \times 6}) + 0.03 \int_0^6 e^{-0.056t} - e^{-0.056t - 0.006t} dt \\
 &= 0.1325 + \frac{0.03}{0.056} (1 - e^{-0.056 \times 6}) - 0.03 e^{-0.3} \int_0^6 e^{-0.006t} dt \\
 &= 0.1325 + 0.1529 - \frac{0.03}{0.006} e^{-0.3} (1 - e^{-0.006 \times 6}) \\
 &= 0.1325 + 0.1529 - 0.1310 = 0.1544
 \end{aligned}$$

$$0.1544 + 0.1310 + 0.7146 = 1 \checkmark$$

$$\begin{pmatrix} 0.7146 & 0.1310 & 0.1544 \\ 0 & 0.7408 & 0.2592 \\ 0 & 0 & 1 \end{pmatrix}$$

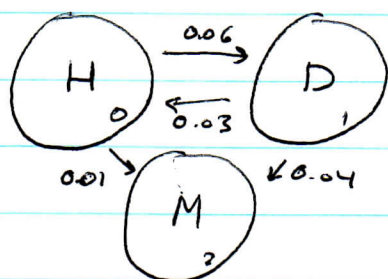
3.



went: $3.5 P_x^{IA}$

$$\begin{aligned}
 &= \int_0^{3.5} e^{-(0.01+0.02)t} (0.01) e^{-(0.3+0.4)(2.5-t)} dt \\
 &= \int_0^{3.5} (e^{-0.03t}) (0.01) (e^{-0.7(2.5-t)}) dt \\
 &= 0.01 \int_0^{3.5} e^{-2.45+0.67t} dt = 0.01 e^{-2.45} \int_0^{3.5} e^{0.67t} dt \\
 &= 0.01 e^{-2.45} (e^{0.67 \times 3.5} - 1) = \boxed{0.00814 = 3.5 P_x^{01}}
 \end{aligned}$$

4.



a.

$$(i) \frac{d}{dt} {}_tP_x^{11} = \sum_{\substack{k=0 \\ k+j=1}}^2 \left({}_tP_x^{10} \mu_{x+t}^{01} - {}_tP_x^{11} \mu_{x+t}^{10} \right) + \left({}_tP_x^{12} \mu_{x+t}^{21} - {}_tP_x^{11} \mu_{x+t}^{12} \right)$$

$$\boxed{\frac{d}{dt} {}_tP_x^{11} = {}_tP_x^{10} \mu_{x+t}^{01} - {}_tP_x^{11} \mu_{x+t}^{10} - {}_tP_x^{11} \mu_{x+t}^{12}}$$

$$(ii) \boxed{\frac{d}{dt} {}_tP_x^{10} = {}_tP_x^{11} \mu_{x+t}^{10} - {}_tP_x^{10} \mu_{x+t}^{01} - {}_tP_x^{10} \mu_{x+t}^{02}}$$

$$(iii) \boxed{\frac{d}{dt} {}_tP_x^{12} = {}_tP_x^{10} \mu_{x+t}^{02} + {}_tP_x^{11} \mu_{x+t}^{12}}$$

b. (x) in l. $\mu_{x+t} \rightarrow 0$ in $t = \frac{1}{2}$. $h = \frac{1}{4}$.

$$\frac{1}{4} P_x^{10} = \cancel{0} + \frac{1}{4} ((1)(0.03) - 0 - 0) = 0.0075$$

$$\frac{1}{4} P_x^{11} = 1 + \frac{1}{4} (0 - (1)(0.03) - (1)(0.04)) = 1 - 0.0175 = 0.9825$$

$$\begin{aligned} \frac{1}{2} P_x^{10} &= 0.0075 + \frac{1}{4} ((0.9825)(0.03) - (0.0075)(0.06) - (0.0075)(0.01)) \\ &= 0.0075 + \frac{1}{4} (0.029475 - 0.00045 - 0.000075) \\ &= \boxed{0.0147375 \approx \frac{1}{2} P_x^{10}} \end{aligned}$$